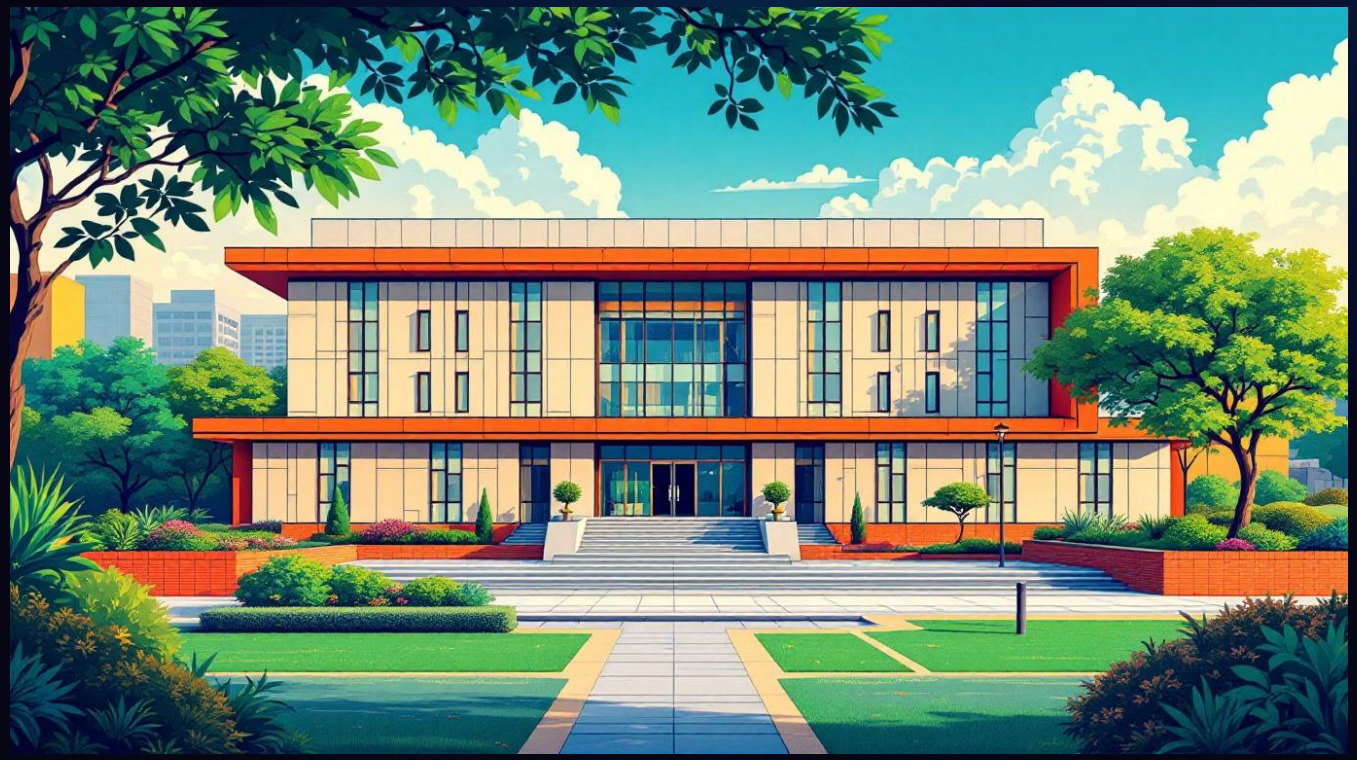
AI Study Planner

A CLI-Based Intelligent Time Management System for Academic Success

Project Overview



Project Details

- **Project Title:** AI Study Planner (CLI-Based)
- **Submitted By:** Pranay Kothari
- **Registration Number:** 25BCE10248
- **University:** VIT Bhopal University
- **Course:** B.Tech Computer Science Engineering

Introduction

Time Management Challenge

Students face significant challenges managing study time across multiple subjects with varying difficulty levels and exam deadlines.

AI Solution

Automated decision-making using logical priority calculation based on weighted parameters rather than manual judgment.

Core Concept

Introduces basic AI principles where decisions are made through heuristic evaluation of subject parameters.

The system addresses the common struggle of allocating appropriate time to each subject when preparing for multiple exams simultaneously.

Problem Statement and Objectives

Problem Statement

Students consistently face challenges in effective study planning:

- Incorrect subject prioritization leading to poor exam performance
- Excessive time spent on less important topics
- Neglect of urgent subjects with approaching deadlines
- Absence of structured, data-driven planning methodology

Project Objectives

- Develop an efficient CLI-based study planner
- Implement intelligent priority calculation algorithm
- Optimize study time allocation across subjects
- Demonstrate practical AI-based decision logic
- Improve overall student productivity and exam readiness

Methodology and System Flow

01

Collect Input

User provides number of subjects and details including difficulty, importance, and days remaining

02

Calculate Priority

Apply weighted formula to determine subject priority scores

03

Sort Subjects

Arrange subjects in descending order of priority

04

Allocate Time

Distribute available study hours proportionally based on priority

05

Display Schedule

Present optimized study timetable to user

Mathematical Model: Core AI Logic

Weighted Formula

The system employs a heuristic evaluation function:

$$Priority = (D \times 0.4) + (I \times 0.4) + \left(\frac{1}{T} \times 0.2\right)$$

Where **D** = Difficulty, **I** = Importance, **T** = Days Left

Parameter Explanation

Difficulty (40%)

Higher values indicate greater effort and preparation time required

Importance (40%)

Higher values reflect greater exam weightage and academic significance

Urgency (20%)

Less time remaining creates higher urgency through inverse relationship

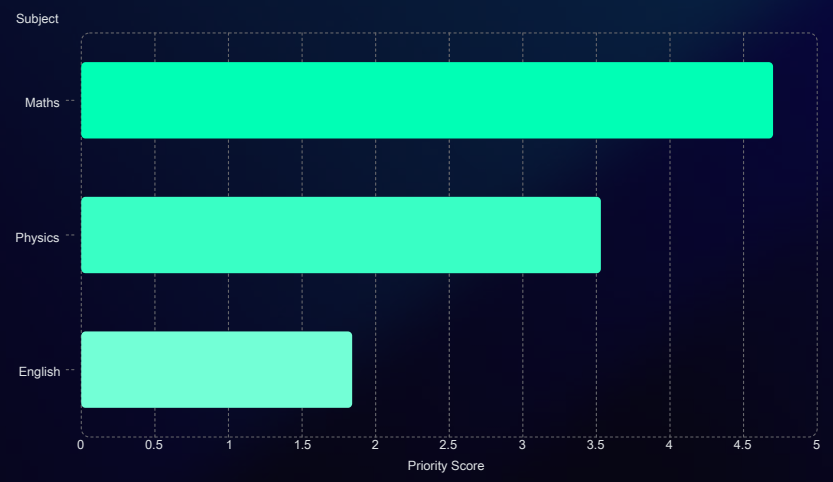
This model simulates intelligent decision-making by combining multiple factors into a single prioritization metric.

Example Analysis and Results

Sample Input Data

Subject	Difficulty	Importance	Days Left
Maths	5	5	2
Physics	4	4	3
English	2	3	5

Priority Comparison



2.8

Maths Study Hours

Allocated from 6 total hours

2.1

Physics Study Hours

Second priority allocation

1.1

English Study Hours

Minimum required time

Implementation Details

Python 3

Core programming language utilizing modern syntax and standard library features for efficient implementation

Object-Oriented Programming

Subject class encapsulates data and methods, promoting code reusability and maintainability

CLI Interface

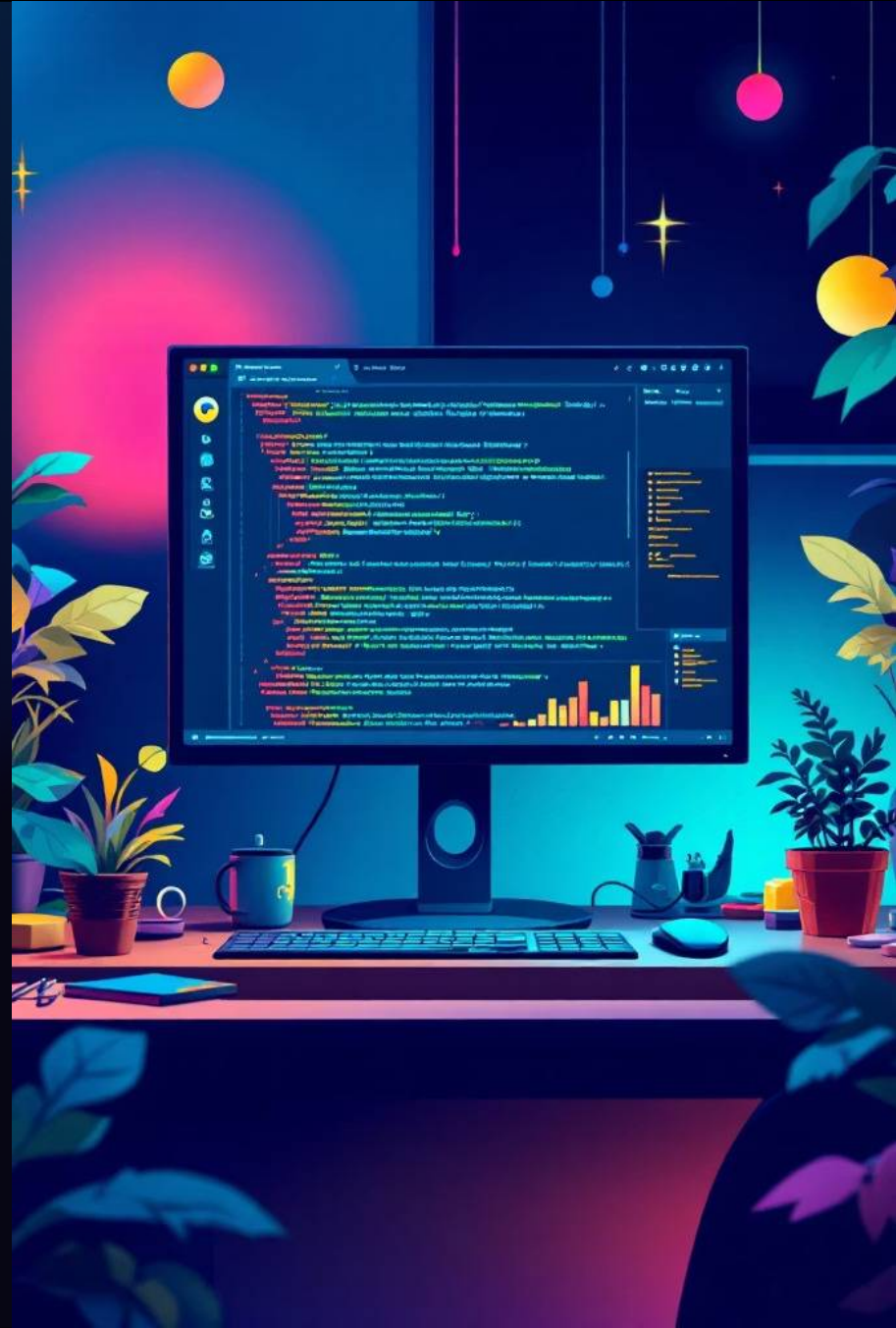
Command-line interface provides direct user interaction with clear input prompts and formatted output

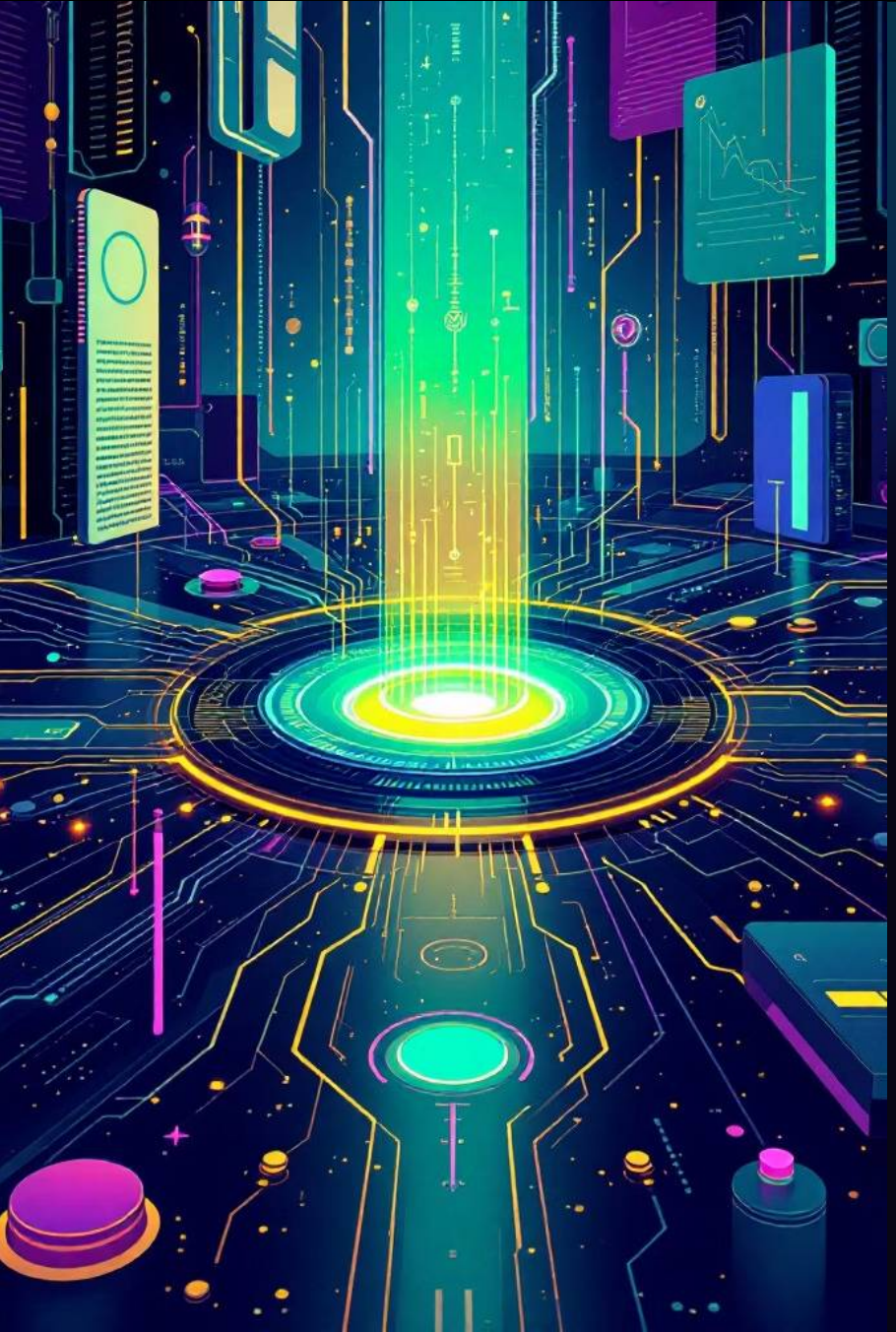
Key Components

- Subject class with attributes and methods
- Priority calculation function with weighted formula
- Scheduling algorithm for time distribution
- Input validation and error handling

Code Logic Summary

- Objects store subject-specific data
- Automatic priority calculation on object creation
- Built-in sorting ensures correct order
- Proportional time distribution based on priorities





Advantages, Limitations, and Future Scope

7 Advantages

- Simple, efficient design with minimal complexity
- Easy to use with intuitive CLI interface
- No external dependencies required
- Logical, data-driven decision-making
- Measurably improves productivity

Limitations

- No graphical user interface
- No persistent data storage capability
- Static planning without adjustments
- Limited personalization features

± Future Enhancements

- GUI interface with Tkinter or web framework
- AI/ML-based adaptive recommendations
- Weekly planner with recurring schedules
- Performance tracking integration
- Mobile application development

Conclusion

Successful Demonstration

The AI Study Planner effectively illustrates how fundamental AI principles can optimize daily study schedules through intelligent parameter evaluation.

Foundation for Future Systems

This project establishes a solid foundation for more advanced intelligent planning systems and highlights AI's practical value in everyday problem-solving.

Real-World Impact

By combining multiple factors into a unified decision-making model, the system ensures effective and balanced time allocation for improved academic performance.